IPV6 CONFIGURATION AND SUBNETTING

Assignment

Name: _____ Class: _____

Date: _____

NETWORK

TASK 1

QUESTION

Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

ANSWER

Note:

This is a sample procedure. Replace the example interface name and IPv6 prefix with details suitable for your own LAN. Do not copy an address that conflicts with another device. A static address may interrupt connectivity if the prefix or gateway is incorrect.

Windows (PowerShell as Administrator):

1. Identify the adapter: Get-NetAdapter

2. Add a sample static IPv6 address (replace Ethernet with your adapter name): New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress "2001:db8:1::25" -PrefixLength 64

3. Verify: ipconfig /all

4. Check the IPv6 address under the correct adapter. The 2001:db8::/32 range is reserved for documentation and is not a real, routable public address; use your network's assigned prefix for an actual configuration.

Linux (example):

1. Find the interface name and current addresses: ip -6 addr show

2. Temporarily add an example address (replace eth0 with your interface and use a prefix valid on your LAN): sudo ip -6 addr add 2001:db8:1::25/64 dev eth0

3. Verify: ip -6 addr show dev eth0

4. This ip command change is temporary and normally disappears after reboot or interface/network reconfiguration.

Expected result:

The selected adapter should show the IPv6 address and /64 prefix. The example documentation address is for practice only; for real LAN connectivity, use a prefix actually configured on your router/LAN.

NETWORK

TASK 2

QUESTION

Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

ANSWER

Steps:

1. Choose a unique-local prefix for the LAN, such as fd12:3456:789a:1::/64. Ensure the other device uses the same /64 prefix and a different host address.

2. On Linux, add a temporary address: sudo ip -6 addr add fd12:3456:789a:1::10/64 dev eth0 (replace eth0 with your interface).

3. On the second LAN device, configure fd12:3456:789a:1::11/64 on its connected interface.

4. Test from the first Linux device: ping -6 fd12:3456:789a:1::11

5. On Windows, inspect addresses with ipconfig and test using: ping -6 fd12:3456:789a:1::11

Windows sample:

In an elevated PowerShell, find the interface index with Get-NetIPInterface -AddressFamily IPv6. Add the sample address with: New-NetIPAddress -InterfaceIndex <index> -IPAddress fd12:3456:789a:1::10 -PrefixLength 64. Verify with ipconfig /all. Remove the example later if it is not needed.

Expected result:

Successful replies indicate IPv6 communication between the two devices. If ping fails, check that both devices share the same LAN/VLAN and /64 prefix, the addresses are unique, IPv6 is enabled, and the firewall permits ICMPv6 echo requests.

NETWORK

TASK 3

QUESTION

Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Document the steps and observed address behavior in a text file.

ANSWER

Steps (Linux example):

1. Record the interface name and current IPv6 addresses: ip -6 addr show

2. Add a temporary address: sudo ip -6 addr add fd12:3456:789a:1::50/64 dev eth0 (replace eth0 with the correct interface).

3. Confirm it appears: ip -6 addr show dev eth0

4. Disconnect/reconnect the interface, then run the same command. Record whether the address remains.

5. Reboot and run ip -6 addr show again. A manually added address from the ip command normally does not persist across reboot.

Sample observation to put in report:

Before reboot: fd12:3456:789a:1::50/64 was visible on the selected interface. After network reconnect: it may remain or be removed depending on how the interface/network is managed. After reboot: the manually added address was absent because it was configured temporarily. Replace this sample with your actual observations.

Windows note:

A temporary Windows address can be added using New-NetIPAddress in elevated PowerShell, then checked with ipconfig /all. Record actual behavior after disconnect/reconnect and reboot; behavior depends on the configuration method and network management.

NETWORK

TASK 4

QUESTION

Simulate a Zomato-style search: Write a small script that stores an array of restaurant names and their IPv6 addresses, then lets the user input a partial name and returns the matching restaurant's IPv6 address.

ANSWER

Python script:

```
restaurants = [
    {"name": "Spice Garden", "ipv6": "fd12:3456:789a:1::101"},
    {"name": "Tandoori House", "ipv6": "fd12:3456:789a:1::102"},
    {"name": "Green Bowl", "ipv6": "fd12:3456:789a:1::103"},
    {"name": "Pizza Corner", "ipv6": "fd12:3456:789a:1::104"}
]

search = input("Enter part of a restaurant name: ").strip().lower()
matches = [r for r in restaurants if search in r["name"].lower()]

if matches:
    for restaurant in matches:
        print(f'{restaurant["name"]}: {restaurant["ipv6"]}')
else:
    print("No matching restaurant found.")
```

How it works:

The script stores restaurant names and example unique-local IPv6 addresses in a list. It converts the user's search text and each restaurant name to lowercase, so the search is not case-sensitive. It prints every matching name and address, or a message if no match is found.

Note:

These fd12:... addresses are examples for a private lab only; they do not identify real restaurants or public internet services.

NETWORK

TASK 5

QUESTION

Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario with at least three subnets, then manually assign a valid IPv6 address from one subnet to your system. Paste the AI-generated scenario and configuration steps in a text file.

ANSWER

AI-generated scenario:

A small delivery startup has one IPv6 unique-local block, fd12:3456:789a::/48. Divide it into three /64 subnets: Delivery Partners = fd12:3456:789a:1::/64; Restaurants = fd12:3456:789a:2::/64; Admins = fd12:3456:789a:3::/64. Example host addresses are fd12:3456:789a:1::10, fd12:3456:789a:2::10, and fd12:3456:789a:3::10. Each group should use the matching /64 prefix and a unique host address.

Linux configuration example:

1. Identify interface: ip -6 addr show

2. Add an address from the Delivery Partners subnet (replace eth0 with your interface): sudo ip -6 addr add fd12:3456:789a:1::10/64 dev eth0

3. Verify: ip -6 addr show dev eth0

4. If needed, remove it after the lab: sudo ip -6 addr del fd12:3456:789a:1::10/64 dev eth0

Windows configuration example:

1. Open PowerShell as Administrator and identify the interface index: Get-NetIPInterface -AddressFamily IPv6

2. Add an example address: New-NetIPAddress -InterfaceIndex <index> -IPAddress fd12:3456:789a:1::10 -PrefixLength 64

3. Verify: ipconfig /all

4. Remove it after the lab if needed: Remove-NetIPAddress -InterfaceIndex <index> -IPAddress fd12:3456:789a:1::10 -Confirm:$false

Important:

Use a prefix that is appropriate for your lab network and ensure it does not conflict with an existing address. Because these are unique-local addresses, they are intended for private networks and are not globally routable. A real connection to another device requires the other device and LAN to be configured consistently.

NETWORK